

Nettonåverdi, NNV og Nåverdiskvote

Nettonåverdi, NNV

- Teoretisk overlegen metode
- Nettonåverdi er summen av alle nåverdi:

$$NNV = k_0 + \sum_{t=1}^{N_e} \frac{k_t}{(1+r)^t}$$

$$NNV = -G + \sum_{t=1}^{N_e} \frac{a_t}{(1+r)^t} + \frac{S}{(1+r)^{N_e}}$$

- Jo høyere nettonåverdi jo bedre
- Invester hvis nettonåverdien er positiv

Snarvei - nåverdisum

$$\sum_{k=1}^{N_e} \frac{a}{(1+r)^k} = a \left(\frac{1}{(1+r)^1} + \frac{1}{(1+r)^2} + \frac{1}{(1+r)^3} + \dots + \frac{1}{(1+r)^{N_e}} \right)$$

Sett x til: $x = 1/(1+r) \Leftrightarrow 1/x = (1+r) \Rightarrow$

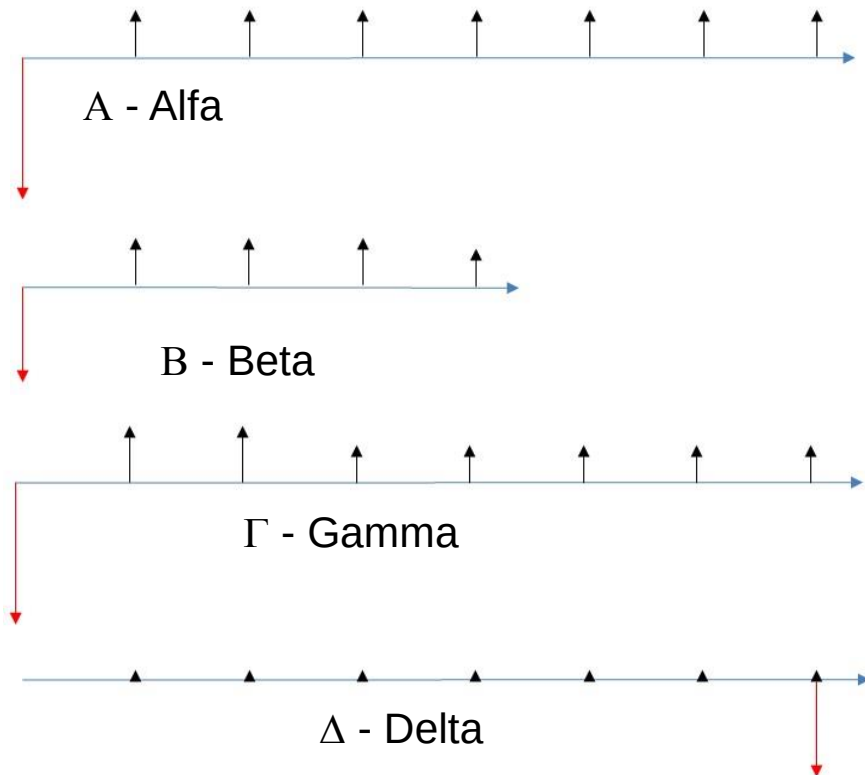
$$\frac{1}{(1+r)^1} + \dots + \frac{1}{(1+r)^{N_e}} = x + x^2 + x^3 + \dots + x^{N_e}$$

$$\frac{(1-x) \cdot (x + x^2 + x^3 + \dots + x^{N_e})}{1-x} = \frac{x - x^2 + x^2 - x^3 + \dots - x^{N_e+1}}{1-x} =$$

$$\frac{x - x^{N_e+1}}{1-x} = \frac{x/x - x^{N_e+1}/x}{1/x - x/x} = \frac{1 - x^{N_e}}{1/x - 1} = \frac{1 - (1+r)^{-N_e}}{1+r - 1} = \frac{1 - (1+r)^{-N_e}}{r}$$

Eksempel

	A	B	Γ	Δ
G	150	100	150	1
N_e	7	4	7	7
a_1	50	50	60	5
a_2	50	50	60	5
a_3	50	50	40	5
a_4	50	40	40	5
a_5	50		40	5
a_6	50		40	5
a_7	50		40	5
S	0	0	0	-100
r	20%	20%	20%	20%



Nettonåverdi - Eksempel

	A	B	Γ	Δ
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a_6	50		40	5
a_7	50		40	5
S	0	0	0	-100
r	20%	20%	20%	20%

$$NNV = k_0 + \sum_{t=1}^{N_e} \frac{k_t}{(1+r)^t} =$$

$$-G + \sum_{t=1}^{N_e} \frac{a_t}{(1+r)^t} + \frac{S}{(1+r)^{N_e}}$$

$$NNV_A = -150 + 50 \cdot \frac{1 - 1.2^{-7}}{0.2} = \mathbf{30.23}$$

$$NNV_B = -100 + \frac{50}{1.2} + \frac{50}{1.2^2} + \frac{50}{1.2^3} + \frac{40}{1.2^4} = \mathbf{24.61}$$

$$NNV_{\Gamma} = -150 + \frac{60}{1.2} + \frac{60}{1.2^2} + \frac{40}{1.2^2} \cdot \frac{1 - 1.2^{-5}}{0.2} = \mathbf{24.74}$$

$$NNV_{\Delta} = -1 + 5 \cdot \frac{1 - 1.2^{-7}}{0.2} + \frac{-100}{1.2^7} = \mathbf{-10.89}$$

Nåverdiskvoten

- Nåverdiskvoten = $\frac{NNV}{G}$
- Brukes ved kapitalmangel
- Tilnærmet "Most bang for the buck"

$$\frac{NNV_A}{G_A} = \frac{30.23}{150} = \mathbf{0.21}$$

$$\frac{NNV_\Gamma}{G_\Gamma} = \frac{24.74}{150} = \mathbf{0.16}$$

$$\frac{NNV_B}{G_B} = \frac{24.61}{100} = \mathbf{0.25}$$

$$\frac{NNV_\Delta}{G_\Delta} = -\frac{10.89}{1} = \mathbf{-10.89}$$